

What's in a Name?



The Effect of Judge-Defendant Name Similarity

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Motivation

- Evidence on in-group bias relies on a narrow set of observables (race, gender, religion)
- Judges observe much more information about defendants than researchers
- Existing measures could be missing key in-group dimensions and leave out social-distance confounders

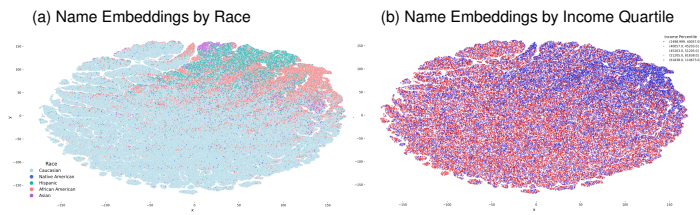
How does social distance affect in-group bias?

- Measure social distance by looking at judge-defendant name similarity
- Investigate in-group bias in Wisconsin Criminal Courts

Data

- More than 1M of criminal cases from 2005 to 2017 in Wisconsin
 - Defendants demographics (race, gender, address)
 - Case information (charges, sentencing, prosecutor and judge, etc.)
 - Matched with hand-collected data on judges' biographies
- Compute FastText embeddings for defendants living in the same street

Figure 1: Two-Dimensional Plot Name Embeddings



Notes: The figure plots the distribution of the name embeddings in the 2-dimensional space, where the two dimensions are computed using TSNE. Panel (a) shows the distribution by defendants' race, while Panel (b) shows the distribution by zip code income quartile.

Judge-Defendant Name Similarity

- Apply name embeddings model to judges' names
- Remove semantic noise from embeddings via demeaning
- For each case with defendant i and judge j calculate

$$\text{NameSimilarity}_{ij} = \text{sim}(\text{NameEmb}_i, \text{NameEmb}_j)$$

What does Name Similarity capture?

Table 1: Name Similarity and other In-Group Dimensions

	Defendant-Judge Name Similarity				
	(1)	(2)	(3)	(4)	(5)
Black Jud. × Black Def.	0.188** (0.0625)			0.189** (0.0620)	0.191** (0.0639)
Female Jud. × Female Def.		0.0460** (0.00711)		0.0449** (0.00710)	0.0471** (0.00725)
Old Def. × Old Jud.			-0.00858* (0.00386)	-0.00802* (0.00397)	-0.00669* (0.00396)
Obs.	1022103	1022103	1022103	1022103	987737
R ²	0.0124	0.0127	0.0111	0.0146	0.0277
County-Year FE	X	X	X	X	X
Zip Code FE					X

Table 2: Judges and Most Similar Defendants

Judge	Background	Defendant	Name Sim.	Background
Carl Ashley	Black, working-class	Equantis Beeks	0.999	Black, lower-class
		Vinetta Simple	0.998	Black, working-class
Jennifer Dorow	White, middle/upper-class	Luke Wierzchowski	0.956	White, upper-class
		Kristen Gravatt	0.946	White, upper/middle-class

Name similarity correlates with race and gender, but also with other characteristics, like a similar social background

Identification Strategy: Estimating In-Group Bias

$$y_{ijct} = \beta \text{NameSimilarity}_{ij} + \alpha_{ct} + \alpha_i^S + \alpha_j + X'_{ijct}\delta + \varepsilon_{ijct}$$

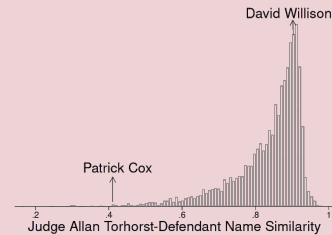
- α_{ct} = county × year fixed effects, α_i^S = charge severity fixed effects:
 - Leverage (conditional) random assignment of judges to cases
- α_j = judge fixed effects:
 - Adjust for all judge characteristics that are constant across cases
- X_{ijct} includes additional defendant, judge, and case characteristics.
 - E.g., flexible controls for defendant recidivism risk (ventile FE).
- Clustering by county-year.

Name-Similarity Driven In-Group Bias

Table 2: In-Group Bias—Effect of Name Similarity on Judicial Outcomes

	Defendant is Incarcerated			Jail Time (Log Days)		
	(1)	(2)	(3)	(4)	(5)	(6)
Name Similarity	-0.014* (0.006)	-0.029** (0.006)	-0.029** (0.006)	-0.130** (0.020)	-0.058** (0.019)	-0.064** (0.019)
Recid. Score			0.061** (0.004)			0.145** (0.015)
Name Similarity × Recid. Score			0.017** (0.005)			0.053** (0.018)
Obs.	1022083	1008865	1008865	397010	391986	3919860
R ²	0.104	0.127	0.126	0.550	0.566	0.566
Average Outcome	.393	.393	.393	1.848	1.848	1.848
SD Name Similarity	0.148	0.148	0.148	0.147	0.147	0.147
$\beta_{\text{NameSim}} / \beta_{\text{BlackJud}} \times \text{BlackDef}$	-0.298	-0.717	-0.713			
$\beta_{\text{NameSim}} / \beta_{\text{BlackDef}}$				-1.073	0.378	0.410
County-Year FE	X	X	X	X	X	X
Charge Severity FE	X	X	X	X	X	X
Risk Ventile FE	X	X	X	X	X	X
Judge FE	X	X	X	X	X	X
Additional Controls		X	X		X	X

Notes: Effect of judges and defendants having a high name similarity on judicial outcomes. Columns 1 to 3 report the results for incarceration decisions, while Columns 4 to 6 report the results for sentence length calculated as the inverse hyperbolic sine of the number of days. Columns 3 and 6 also report the results on the heterogeneity of the name similarity effect for increasing defendants' recidivism risk score, computed using a machine learning model. All columns include fixed effects for county-year, charge severity, and judges. Columns 1, 2, 4, and 5 also include recidivism risk ventiles fixed effects; Columns 2, 3, 5, and 6 also control for other judge and defendant characteristics and other in-group dimensions. The table also reports the standard deviation of the name similarity score and the ratio between the coefficients on the name similarity and the racial in-group bias coefficient (Columns 1 to 3) and the racial disparities one (Column 4 to 6) to provide a benchmark for the effect. Standard errors (in parentheses) are clustered at the county-year level: * $p < 0.1$, * $p < 0.05$, ** $p < 0.01$.



- Holding constant gender, race, and charge severity, an increase of 0.5 in the similarity decreases the likelihood of being incarcerated by 1.5 percentage points
- Robust to including zip code FE
- Not driven by implicit egoism (Chen, 2019)
- No effect of sharing the first name

Heterogeneity by Defendants' Predicted Recidivism Risk

Model of judicial decision making where judges can be from the same group as defendants ($m_{ij} = 1$) and observe a signal about defendants' true recidivism risk ($\tilde{r}_{ij} = r_i + \epsilon_{ij}$, $r_i \sim \mathbf{N}(\mu_i, 1)$ and $\epsilon_{ij} \sim \mathbf{N}(0, \frac{1}{\rho_{ij}})$). Judges incarcerate if

$$\frac{\phi m_{ij}}{\text{In-Group Bias}} - \frac{\tau \tilde{r}_{ij}}{\text{Cost of Recidivism}} < \frac{\kappa_0}{\text{Cost of Incarcerating}}$$

Assuming signal precision increases if judges and defendants are from the same group $\rho_{ij} = \hat{\rho} + \rho_m m_{ij}$, then

$$\frac{\partial^2 U(\text{jail})}{\partial \tilde{r}_{ij} \partial m_{ij}} = \tau \frac{\rho_m}{(1 + \rho(\cdot))^2} > 0$$

With higher type similarity, judges get higher expected utility from incarcerating higher-risk defendants, as we observe in Table 2.

- Results are not driven by the introduction of COMPAS
- Higher responsiveness to recidivism risk score only when this is trained using criminal history

Conclusion

- Evidence on a new dimension of in-group bias: **Social Distance**
 - Step forward in understanding drivers of judicial decision-making, beyond racial and gender considerations
 - More responsiveness to recidivism risk for judges-defendants pairs with lower social distance
- Results could be helpful in the design of policies addressing criminal justice disparities